

1. (a) (i) For all $x \neq \frac{1}{2}$, $X_n(x) = 0$ for large n , and $P(\{\frac{1}{2}\}) = 0$. This proves $X_n \rightarrow X$ almost surely (P) as $n \rightarrow \infty$.
- (ii) Since for fixed $T > 0$, $\int |X_n| I_{\{|X_n| > T\}} dP = n.m([\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}]) = n \cdot \frac{2}{n} = 2$, $\forall n > T$, $\{X_n\}$ is not UI.
- (iii) Here $E|X_n - X| = E(X_n) = n \cdot \frac{2}{n} = 2$ for all n large, so X_n does not converge to X in L^1 .
- (b) If $\{X_n^p\}$ is UI, and the measure is finite (which is the case in the example above), then $X_n \rightarrow X$ a.e (P) would imply $X_n \rightarrow X$ in L^p for $p \geq 1$ (HW problem: 2.36(b)). There are other correct answers too, for example, along the lines of DCT.

2. (a) $\mu_3(\emptyset) = 2\mu_1(\emptyset) + 3\mu_2(\emptyset) = 0$;
 $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall A \in \mathcal{F}$;
 If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = 2\mu_1(\cup_i A_i) + 3\mu_2(\cup_i A_i) = 2\sum_i \mu_1(A_i) + 3\sum_i \mu_2(A_i) = \sum_i (2\mu_1(A_i) + 3\mu_2(A_i)) = \sum_i \mu_3(A_i)$.
 So μ_3 is a measure.
- (b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = 2\mu_1(A) + 3\mu_2(A) = 0$. So $\mu_3 \ll \nu$.
- (c) Claim: $\frac{d\mu_3}{d\nu} = 2\frac{d\mu_1}{d\nu} + 3\frac{d\mu_2}{d\nu}$ a.e (ν). *Proof:* Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists f_1, f_2 \geq 0$ s.t $\mu_1(A) = \int_A f_1 d\nu$, $\mu_2(A) = \int_A f_2 d\nu$. By definition, $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A) = 2\int_A f_1 d\nu + 3\int_A f_2 d\nu = \int_A (2f_1 + 3f_2) d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = 2f_1 + 3f_2 = 2\frac{d\mu_1}{d\nu} + 3\frac{d\mu_2}{d\nu}$ a.e (ν).
- (d) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $3\mu_2(B^c) = 0 = \nu(B)$, i.e. $3\mu_2 \perp \nu$. Choose $\mu_s = 3\mu_2$, $\mu_a = 2\mu_1$, then $\mu = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$.
 Note that if $\phi_{2,3}$ represent the density of $Normal(2, 3)$, then by definition $\mu_1(A) = \int_A \phi_{2,3} dm$ and $\nu = m$. So, $\frac{d\mu_1}{d\nu} = \phi_{2,3}$ from RN theorem. Hence, from part (a) above, $\frac{d\mu_a}{d\nu} = \frac{d(2\mu_1)}{d\nu} = 2\frac{d\mu_1}{d\nu} = 2\phi_{2,3}$.

3. (a) Using Markov's inequality, for each $t > 0$,

$$P(X > a) = P(e^{tX} > e^{ta}) \leq e^{-ta} M(t),$$

and hence the result follows.

- (b) Let $\phi(x) = -\log x$, $x > 1$, then $\phi(x)$ is a convex function.
 If $EX < \infty$, then since $0 \leq \log x \leq x$, $\forall x \geq 1$, we have $0 \leq E(\log X) \leq EX$, i.e. $E \log X < \infty$. By Jensen's inequality, $E\phi(X) \geq \phi(EX)$, i.e. $E(\log X) \leq \log(EX)$.
 If $EX = \infty$, then $E \log X \leq \infty = \log(EX)$ holds trivially.
 Hence $E(\log X) \leq \log(EX)$.

4. (a) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e. (\mu), \forall n \geq 1, \int g d\mu < \infty$ and $f_n \rightarrow f$ a.e. (μ) , then $f \in L^1(\Omega, \mathcal{F}, \mu), \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

(b) We need to prove the following modified DCT:

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e. (\mu), \forall n \geq 1, \int g d\mu < \infty$ and $f_n \rightarrow f$ in measure, then $f \in L^1(\Omega, \mathcal{F}, \mu), \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

Proof: First we state two facts in real analysis:

Fact 1: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \rightarrow x$.

Fact 2 (Theorem 2.5.2): $f_n \rightarrow f$ in measure $\Rightarrow$ there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \rightarrow f$ a.e. (μ) .

Then we prove $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$.

Denote $a_n = \int f_n d\mu, a = \int f d\mu$ and $b_n = \int |f_n - f| d\mu$. To show $a_n \rightarrow a$ and $b_n \rightarrow 0$, we need to use Fact 1. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \rightarrow f$ in measure, then $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0, \forall \varepsilon > 0$, thus $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0, \forall \varepsilon > 0$, i.e. $f_{n_k} \rightarrow f$ in measure. By Fact 2, we can get a sub-subsequence $f_{n_{k_l}} \rightarrow f$ a.e. (μ) . For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have

$f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$, i.e. $a_{n_{k_l}} \rightarrow a$. By Fact 1, we get $a_n \rightarrow a$, i.e. $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. Similarly, we have $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.